

Problem Set 3

● Graded

Student

Jinku Jung

Total Points

27 / 30 pts

Question 1

4.15

4 / 4 pts

Part A

+ 0 pts Incorrect solution or no solution

✓ + 1 pt correct number of bits

✓ + 1 pt correct value using the 5:1 compression ratio

Part B

+ 0 pts Incorrect solution or no solution

✓ + 1 pt Calculates the correct number of uncompressed bits

✓ + 1 pt Correct compression ratio

- 1 pt Used wrong Pixel/Conversion but correct calculations and work shown

Question 2

4.19

■ 4 / 4 pts

✓ + 0.5 pts A correct

✓ + 1 pt B correct

✓ + 0.5 pts C correct

✓ + 1 pt D correct

✓ + 1 pt E correct

+ 0.5 pts Half point round up

💬 Nice job!

Question 3

4.25 Truth Table and Boolean Expression

3 / 3 pts

✓ + 2 pts Correct Truth Table

✓ + 1 pt Boolean Expression matches truth table

+ 0 pts No submission or incorrect

Question 4

4.25 Circuit Diagram

2 / 2 pts

+ 0 pts No submission or incorrect

✓ + 1 pt Circuit diagram matches truth table

✓ + 1 pt Labeled wires/inputs (cannot earn this point if diagram is very difficult to read/understand)

- 1 pt Uses boolean gates that take 3 inputs without showing how it's constructed

Question 5

4.26 Truth Table and Boolean Expression

2 / 4 pts

+ 0 pts No submission or completely incorrect

✓ + 2 pts Correct truth table

+ 1 pt Difference Boolean expression matches truth table

+ 1 pt Borrow out Boolean expression matches the truth table

💬 Double check your difference expression! Also, need to show how the segments of the boolean expression are then connected to get full credit for the boolean expression.

Question 6

4.26 Circuit Diagram

2 / 2 pts

+ 0 pts No submission or completely incorrect

✓ + 1 pt Circuit diagram matches truth table

✓ + 1 pt Labels wires/input (cannot earn this point if diagram is very difficult to read/understand)

- 1 pt Uses boolean gates that take 3 inputs without showing how it's constructed

Question 7

5.19

2 / 3 pts

+ 0 pts Missing or all incorrect

✓ + 1 pt Part A

+ 1 pt Part B

✓ + 1 pt Part C

💬 b) There are 2^{18} possible addresses, but 1 byte can be stored in each address. Therefore, the maximum memory size is 2^{18} bytes.

Question 8

5.21 a.

2 / 2 pts

+ 0 pts Missing or incorrect

✓ + 1 pt Correct syntactical structure

✓ + 1 pt Program functions correctly

Question 9

5.21 b.

2 / 2 pts

+ 0 pts Missing or incorrect

✓ + 1 pt Correct syntactical structure

✓ + 1 pt Program functions correctly

Question 10

5.21 c.

2 / 2 pts

+ 0 pts Missing or incorrect

✓ + 1 pt Correct syntactical structure

✓ + 1 pt Program functions correctly

💬 Great job!

Question 11

5.21 d.

2 / 2 pts

✓ + 1 pt Correct syntactical structure

✓ + 1 pt Program functions correctly

+ 0 pts Missing or incorrect

Question 12

Formatting

0 / 0 pts

✓ - 0 pts Good formatting

- 1 pt Formatting: Improper format (anything other than a single PDF or zipped single PDF) or mistagged

Question assigned to the following page: [1](#)

~~4/15/21~~ (b)

180. second song $\left(\frac{40,000 \text{ samples}}{1 \text{ second}} \right) \left(\frac{16 \text{ bits}}{1 \text{ sample}} \right) = 115,200,000$ bits needed to store this song.

★ If a 5:1 compression scheme is used,
 $\frac{115,200,000}{5} = 23,040,000$ bits needed to store this compressed song

(b) RGB $\rightarrow$ 24-bit color depth/pixel

so
 $\left(\begin{array}{l} \text{Total \# of bits} \\ \text{needed to store} \\ 1200 \times 800 \text{ pixel} \\ \text{image} \end{array} \right) = (1200 \times 800) \text{ pixels} \left(\frac{24 \text{ bits}}{1 \text{ pixel}} \right) = 23,040,000$ bits needed to store this image

★ If the image, when compressed, only takes up 2.4 Mbits?
 $\rightarrow$ ★ A 10:1 compression ratio scheme was used!

Question assigned to the following page: [2](#)

ch4/1

Given $a=1$, $b=2$, & $c=2$.

Evaluate:

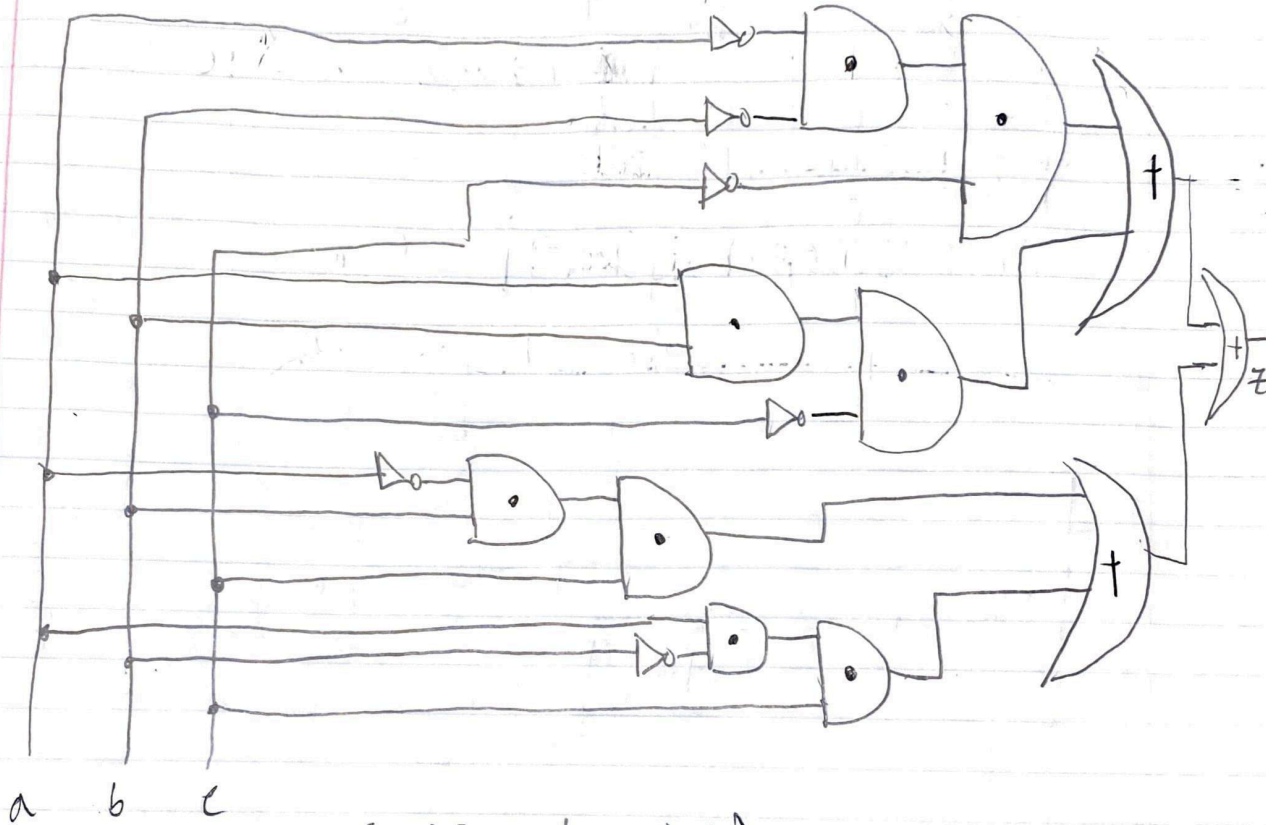
(a) " $(a > 1)$ OR $(b = c)$ " : true(b) " $[(a+b) > c]$ AND $(b < c)$ " : false(c) "NOT $(a=1)$ " : false(d) "NOT $[a=b]$ OR $[b=c]$ " : false(e) " $(a=1)$ AND $(b=1)$ AND $(c=2)$ " : false

Questions assigned to the following page: [3](#) and [4](#)

Ch 4/
 125

Design an "odd-parity" circuit!

$$(a \wedge \neg b \wedge \neg c) \vee (a \wedge b \wedge \neg c) \vee (\neg a \wedge b \wedge c) \vee (a \wedge \neg b \wedge c) \supset \bar{z}$$



Truth Table	(Inputs)			(Output)
	a	b	c	$\bar{z}$
	0	1	1	0
	1	0	1	1
	0	0	1	0
	1	1	0	1
	0	1	0	0
	1	0	0	0
	0	0	0	1

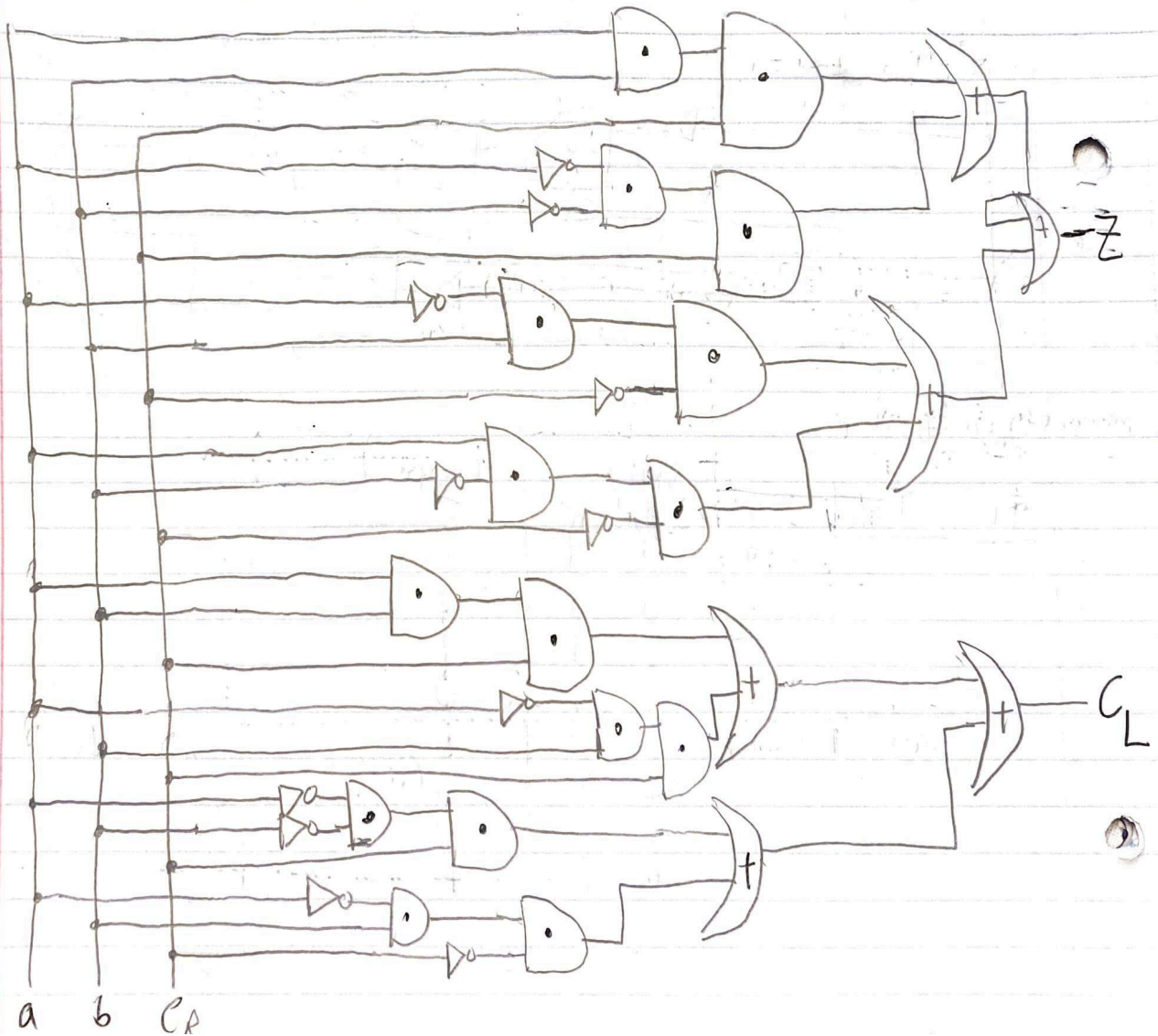
Questions assigned to the following page: [5](#) and [6](#)

PSET #3

Jinkun Jung
 COMS 1004
 HW1: ds 2598
 2/17/2021
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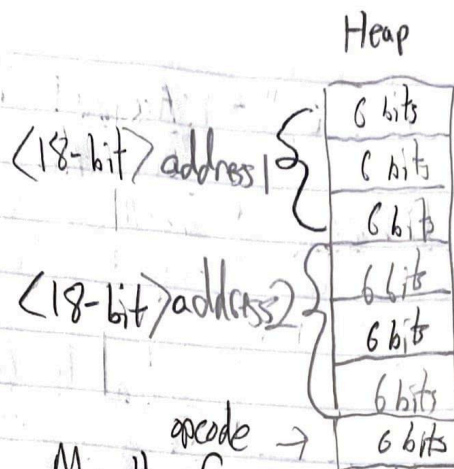
Ch4/ #26

a	b	C_P	C_L	Z (a-b)	For C_L to be true	For "(a-b)" to be true
0	0	0	0	0	$(a \wedge b \wedge C_P)$ $(\neg a \wedge b \wedge C_P)$	$(a \wedge b \wedge C_P)$
0	0	1	0	0	$(\neg a \wedge \neg b \wedge C_P)$	$(\neg a \wedge \neg b \wedge C_P)$
0	1	0	0	0	$(\neg a \wedge b \wedge \neg C_P)$	$(\neg a \wedge b \wedge \neg C_P)$
0	1	1	0	1		
1	0	0	0	1		
1	0	1	0	0		
1	1	0	0	0		
1	1	1	0	0		



Question assigned to the following page: [7](#)

Q5/11 (a)



each op code = 6 bits of memory

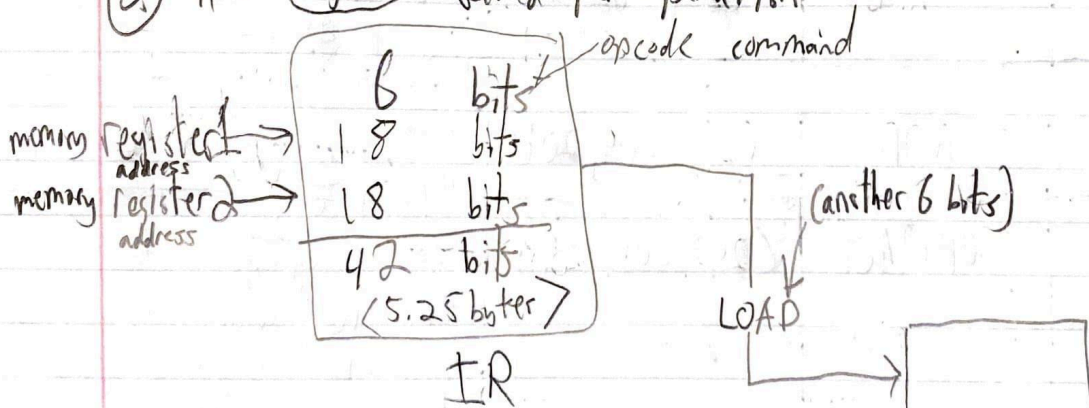
Max # of unique operation codes possible? $2^6 = 64$

"padding" memory cell to take this operation up to the nearest byte value, 6 bytes

(b) Max memory size?

$$2^{18} = 262,144 \text{ bits} = 32,768 \text{ bytes} = 32 \text{ KB}$$

(c) # of bytes required per operation?



So 42 bits + 6 bits = 48 bits (1 byte "x")

= 6 bytes are required per operation

Questions assigned to the following page: [8](#) and [9](#)

For #21

(a) thru
(d) :

variable	memory address in ML
V	200
W	201
X	202
Y	203
Z	204

PSET #3

Jinku Jung
COMS 1004

UID: 2598

2/17/2021

Initial Variable Conditions

ch. 5

(2)

Algorithms

Machine Language Command Equivalents

(a) Set v to the value of $x - y + z$.

* Assume The command "SUBTRACT X, Y, Z" exists in ML, executing the expression "CONC" = $CON(X) - CON(Y)$.

Actual Function in the Command

Sets v equal to " $(x - y)$ "

Adds z to the value previously assigned to v such that:
 $v = (x - y) + z$

Heap Address	Op Code	Heap Cells Used
① 50	SUBTRACT	(202, 203, 200)
② 51	ADD	(200, 204, 200)

(b) Set v to the value of $(w + x) - (y + z)$.

Heap Address	Op Code	Heap Cells Used
① 50	ADD	(201, 202, 200)
② 51	ADD	(203, 204, 201)
③ 52	SUBTRACT	(200, 201, 200)

Actual Function in the Command Sequence
Sets v equal to the expression " $(w + x)$ ".

Sets w equal to the expression " $(y + z)$ ".

Sets v equal to the expression " $(v - w)$ " which is equivalent to our original " $(w + x) - (y + z)$ " difference expression that we wanted based upon our initial variable conditions (see above at top of page).

Question assigned to the following page: [10](#)

< Ch. 5 / #21 cont'd >

① " If $(v \geq w)$ then
 set x to y

Else

 set x to z.

Heap Address	Op Code	Heap Cells Used	Actual Function in the Command Sequence?
① 50	COMPARE	(200, 201)	Compares v and w, such that: (a) If $\text{CON}(x) \geq R$, then $\text{GE} = 1$; else, $\text{GE} = 0$. (b) If $\text{CON}(x) \leq R$, then $\text{LE} = 1$; else, $\text{LE} = 0$. (c) If $\text{CON}(x) = R$, then $\text{EQ} = 1$; else, $\text{EQ} = 0$.
② 51	JUMPGE	(54)	If $v \geq w$, jump to cell 54 and run its command.
③ 52	MOVE	(204, 202)	Set x equal to the value in z.
④ 53	JUMP	(55)	Skip to cell 55 and run its command.
⑤ 54	MOVE	(203, 202)	Set x equal to the value in y.
⑥ 55 : (etc)	< Insert next ^{ML} instruction & its parameters (HERE). >		

Question assigned to the following page: [11](#)

<Ch. 5/#21 cont'd>

(d) "While $y < z$ do

Set y to the value $(y + w + z)$

Set z to the value $(z + v)$

END of the loop."

	Heap Address	Op Code	Heap Cells Used	Actual Function in the Command Sequence
(1)	50	COMPARE	(203, 204)	Compare y to z , then set condition codes.
(2)	51	JUMPLT	(53)	If $y < z$, jump to cell 53 and run its command.
(3)	52	JUMP	(57)	Else, jump to the END of the loop at cell 57.
(4)	53	ADD	(203, 201, 203)	y is now set equal to " $(y + w)$ "
(5)	54	ADD	(203, 204, 203)	y is now set equal to " $(y + w + z)$ "
(6)	55	ADD	(204, 200, 204)	z is now set equal to " $(z + v)$ "
(7)	56	JUMP	50	Go back to cell 50 of the while loop.
(8)	57..(etc)	<Insert next ML instruction & its parameters HERE.>		